2 Across distances

This one's a bit obvious. But what we mean by connectivity is not just hooking people's PCs up to a network, but also figuring out how we can do so seamlessly, and not just limited to PCs, but whatever device we're using. It's also about managing our data better, and making public data easily available to the public. When you think connectivity in the future, we're hoping you will be thinking about not just getting on a network, but also being able to seamlessly use it.

All the things we've been making a noise about in the past decade in technology require people to be connected. eGovernance, education, information flow, planning, disaster management, and even your local traffic report need to connect you to the data that happens to be important to you at just that moment. Figuring out a solution to this would not just solve a lot of our problems, but also play a rather large role in enabling some of the innovations we're going to be talking about a little later in this article. This just makes it all the more critical for us to try and achieve.

Solution: **Cumulo-Indus**

A wireless country-wide data network that's cheap (or free) to access, is truly platform independent, is cloud based.

How it could work

If there's one thing that we're never happy with, it's bandwidth. There's always too little of it, and it's always too expensive. We've heard experts tell us that the reason for this is expensive international bandwidth, and how local bandwidth is cheaper. Thankfully, to create a wide area network (WAN) that spans the country, we actually need no international bandwidth whatsoever. Since costs are taken care of, or at least controllable, the next step is to figure out how we could network the country. The simple answer seems to be wireless technologies, since going forth, they seem more reliable than fiber optics or copper. However, we'd also need to solve for today's problems of high costs and lower bandwidths over large distances when using wireless networks. Plus the weather can play spoilsport again, affecting signal quality, as do all those buildings around you.

Assuming we can come up with a wireless standard of our own, or just adopt one for usage, the next, and perhaps biggest challenge is to connect everything. In an article in Digit a few years ago we looked at what the future (2020) would look like. We assumed that every device we owned would be smart, networked, and programmed to communicate the right information at the right time.

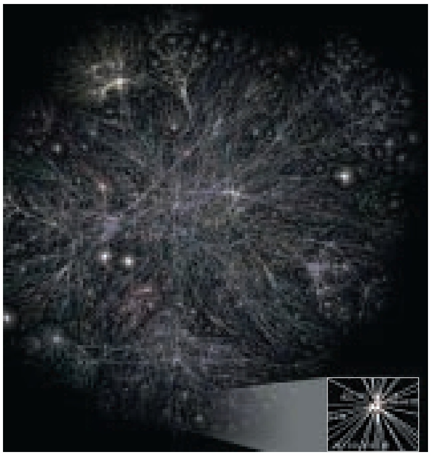
The ordinary

Let's relook at a popular example here. Your fridge notices that it's running low on milk. Now it can just tweet that to the world, or use a phone service to send you an SMS, that you may or may not read. However, imagine the possibilities if the fridge was able to tell that, since you're in a meeting, the reminder would be wasted on you right now, but another family member who happens to be passing a store, at that instant, would be a better target for this information. All this may sound like sci-fi, but it's very programmable even today, it's just the networks that don't exist between a regular fridge and you.

Of course this means you would never run out of milk, but what can it do for India?

The extraordinary

Now imagine that same connectivity applied to everything and everyone.



The net today is complicated enough, imagine the future?

For instance, the truck driver bringing in groceries for your neighbourhood would know better than to turn on to a traffic jammed road. The same would apply if a truck broke down in the middle of a highway, in the dead of night, and was able to call the nearest empty truck for help.

We could also connect all colleges and schools to a grid, enabling content sharing, admission statuses and result announcements. Police stations would get information in real time of arrests or crimes. In case of a flood, the government would be able to tell everyone to get to higher ground...

However, it's not just a question of being connected. Connectivity also opens up the possibilities of database sharing, and access to services. Imagine a farmer in a remote village standing in front of a kiosk in his village, paying for the seeds he's ordered from a distant city, booking the required amount of space on a goods train and also the truck that will carry it all from the nearest station to his farm. Or the reverse. Imagine farmers sending out an alert that they happen to have X amount of grain just lying about, while a vendor in another village agreeing on a price and paying for the goods.

Perhaps we'd also be able to use all our "3G" cell phones, and actually start making video calls. We could also cut down on travel with video calling if we had a reliable network that was cheap to use, thus saving costs and energy.

Of course, if our dream wireless innovation were truly inter-operable across platforms (think Android, Windows, iOS, Linux and more), this would enable devices to do more than just "talk" to each other. The most eagerly awaited innovation would be when devices start to share resources. Imagine your netbook using the coffee maker's and washing machine's IC and memory to speed up displays when you're watching an HD video...

Must-clicks
http://bit.ly/b1fuMd | http://bit.ly/cweeNs
http://bit.ly/cqx4uO